EPSR-test-ex79 Other axis combinations: sharm (x,z / x,-z) and sdfcube (x,y / y,x)

Program(s): sharm, sdfcube | chained passes: 1 | verification: ALL IDENTICAL (1 pass)

What this case tests

Covers the remaining branches of the molecular-axis construction (`form_eulerangles`) used by `sharm` and `sdfcube` [1]: first axis x, second axis z, -z or y, and a single-atom-type centre.

Code exercised

Fortran (original, fortran_src/):

- `sharm_rev2.f`: `run_sharm` (keywords, `nsumt` rule), `sharm` (centres, axes, pair loop, averaged molecules)
- `sdfcube.f`: `run_sdfcube`, `sdfcube` (grid histogram), `initcube`, `outcube` (cube file, accumulation)
- `sharm_routines_rev1.f`: `get_l_and_n_values`, `form_cgcoeffs`, `form_eulerangles`, `form_dlmn`, `form_sum_prod_dlmn_cgcoeff`
- `auxiliary_routines.f`: `dlm`, `cgcoef`, `rotateeuler`; `read_and_save_files.f`: `outgofzs_sharm` (`.h0n` files, averaging); `process_symbols.f`: `check_symbol`

Rust (port, rust/epsr-core/src/):

- `tools/sharm.rs`: `run`, `sharm`, `read_axis`, `read_l_values`, `average_molecule`
- `tools/sdfcube.rs`: `run`, `sdfcube`, `initcube`, `outcube`
- `tools/sharm_routines.rs`: `Sharm` methods with the Fortran names (incl. `outgofzs_sharm`), `dlm`, `cgcoef`, `form_eulerangles`, `rotateeuler`

Physical system

The EPSR-test configuration `200DMSO400Water.ato`: 200 DMSO (atoms S, C, C, O, 6 H) and 400 water molecules (O1, H1, H1; SPC/E parameters [2]), 3200 atoms at 300 K in a cubic box of side 33.12 Å, a model of the DMSO-water mixtures of Soper and Luzar [3]. `sharm` expands the orientational pair correlation function of a central and a second molecule type in rotation matrices coupled by Clebsch-Gordan coefficients; the coefficients g(l1,l2,l;n1,n2;r) are averages over all molecule pairs [4,5]. `axes.SHARM.dat`: water (axes `x 2 3`, `z 2`) and DMSO (axes `x 4`, `-z 2 3`), l1, l2, l = 0-2, n steps 1 (97 coefficients). `sdfcube` histograms the centres of the second molecules, rotated into the frame of each central molecule (Euler angles from the axis definitions), on a cubic grid and writes the spatial density function as a Gaussian `.cube` file together with the averaged molecules (`-first.xyz`, `-second.xyz`). `axes.SDFCUBE.dat`: water (axes `x 2 3`, `y 2`) and DMSO centred on S (axes `y 4`, `x 2 3`), 13×13×13 grid (`mpixel 6`, `rmax 7`), `averagemol y`.

Inputs

Input files written by hand from the keyword tables (`setup_symbols_*`), one `keyword value` per line, ended by `q`; `200DMSO400Water.ato` copied from EPSR-test.

How the accuracy was verified

`commands.txt` (2 program calls, run once) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; no masking except where noted below). The Fortran output is stored in `fortran_reference.tar.gz`. A run-time FORMAT in the original source (`sharm_rev2.f`/`sdf_rev2.f`, and `sharm_diffuse` for `epsr`) lacks a comma after the Hollerith `1h(` and is rejected by current gfortran, so the reference build inserts that comma (originals kept as `*.orig`); the port behaves like the patched program. Result of the final regression run of the complete port: every file identical in 1 pass.
One pass. Compared: `axes.SHARM.h01` (97 coefficients, `nsumt 1`), `axes.SDFCUBE.cube`, the four xyz files and both updated input files. Reference build: `sharm`/`sdf` were rebuilt with one comma inserted in the run-time title format (`'(1h(,'//il1fmt`), since gfortran 15.2 otherwise stops with "Missing comma between descriptors" before writing the coefficients; all other programs are the unmodified reference binaries.

References

[1] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[2] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).
[3] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[4] A. K. Soper, "Orientational correlation function for molecular liquids: The case of liquid water", J. Chem. Phys. 101, 6888-6901 (1994).
[5] C. G. Gray and K. E. Gubbins, Theory of Molecular Fluids, Vol. 1: Fundamentals, Clarendon Press, Oxford (1984).
[6] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.